

BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: ____BIT/2024/74067

School/Faculty: ____COMPUTER AND INFORMATICS

Programme: _____

Academic Year: _____

Semester: _____

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Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

- Installed Android Studio and configured development environment
- Created first Android project (STUDENT PORTALPRO)
- Learned basic structure of Android applications (Activities, XML layouts)
- Ran application on emulator for testing setup

Screenshots/Evidence Attached

only screenshots needed and labeling

Screenshots/Evidence Attached (Yes/No)

Android Studio installation/setup

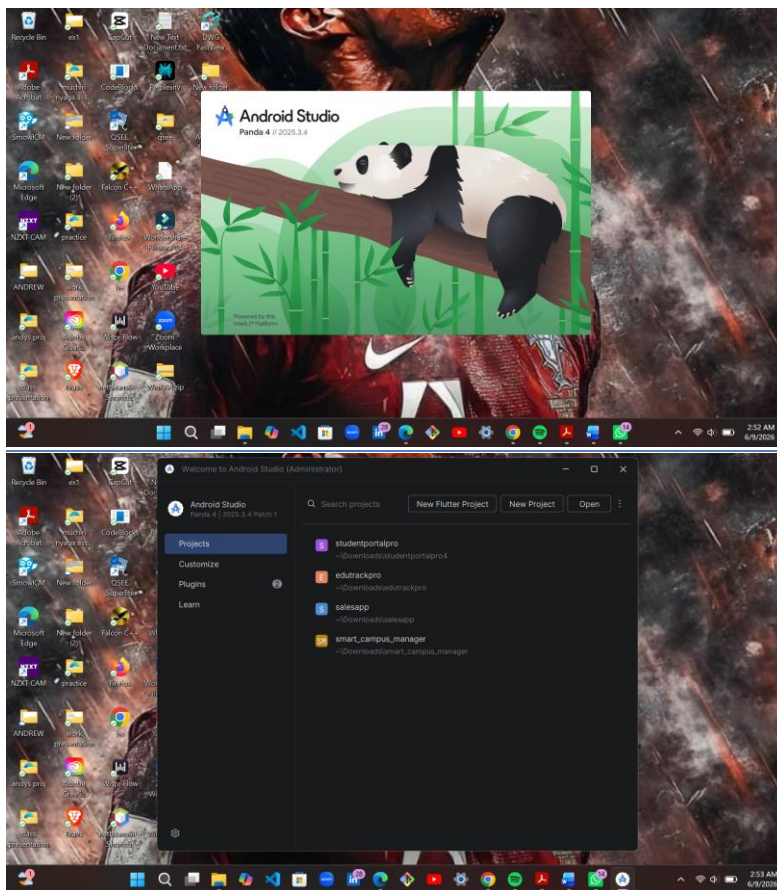


Figure 1. ANDROID STUDIO

INSTALLATIO

Created project structure

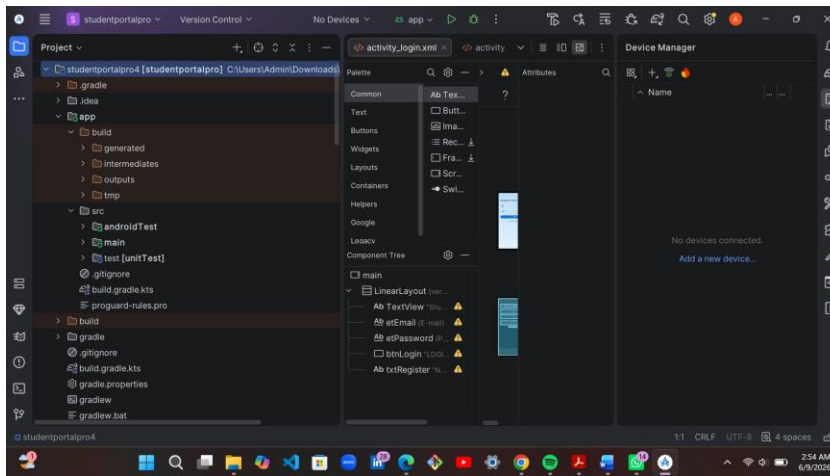


Figure 2.STUDENT PORTAL

PRO PROJECT CREATION

Emulator running

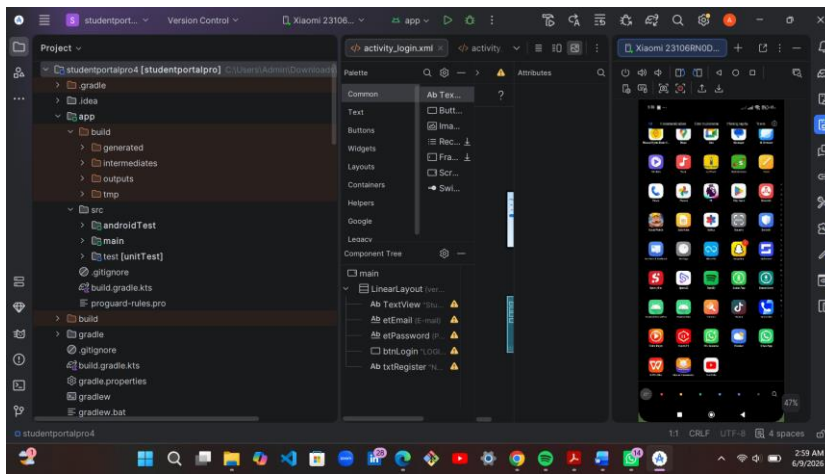


Figure 3.EMULATOR APP

RUNNING

Tools/Software Used

1. Android Studio
2. Java
3. Android Emulator
4. Windows OS

Personal Reflection

This week I learned how to set up Android Studio and create my first mobile application. I understood the basic project structure and how the emulator works for testing apps.

not more than 50 words

Week 2: Mobile Development Environment

Practical Activities Undertaken

- Designed login screen using XML
- Added Edit Text fields for username and password
- Created buttons for login and register
- Applied basic Constraint Layout for alignment

Screenshots/Evidence Attached

Login screen design

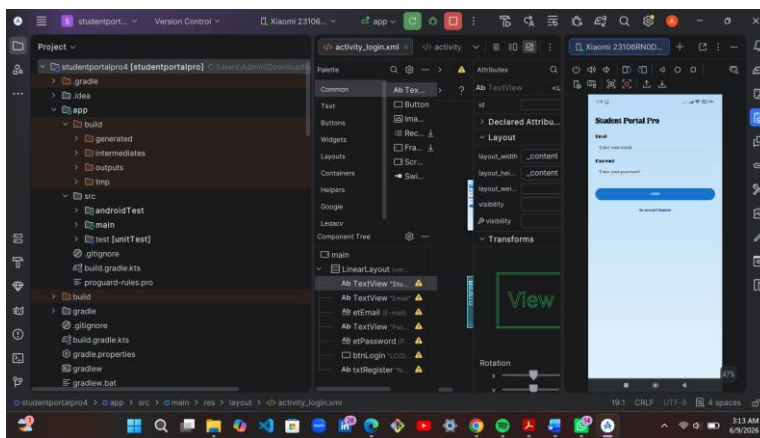
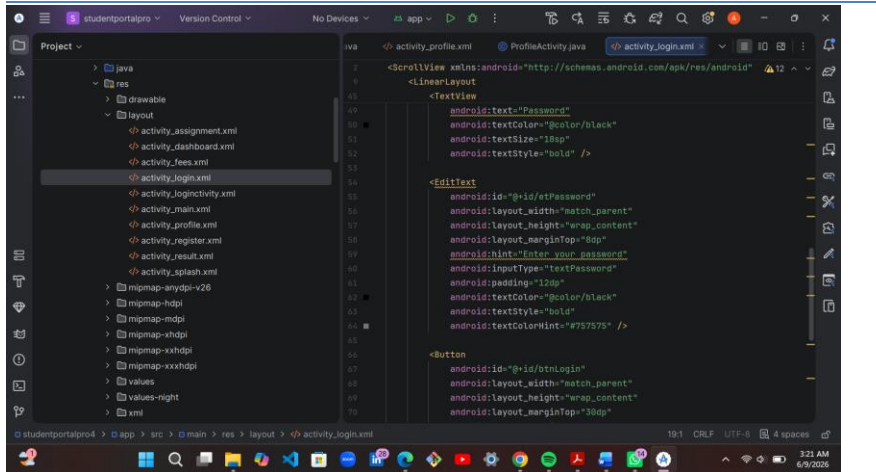
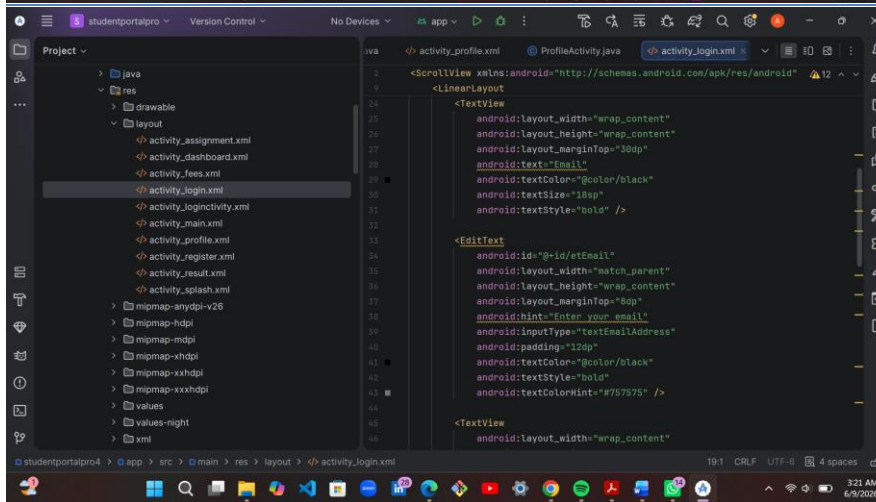
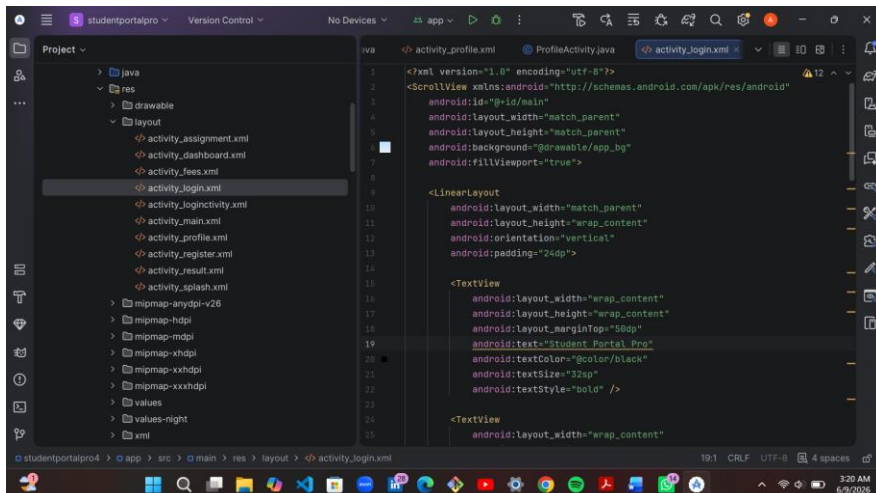


Figure 4.LOGIN SCREENS DESIGN

XML layout code



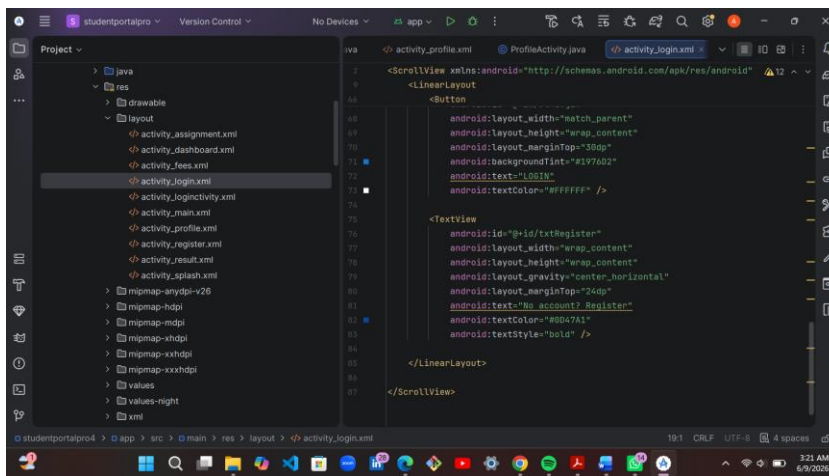


Figure 5.WHOLE

ACTIVITY_LOGIN.XML_CODE

Running UI on emulator

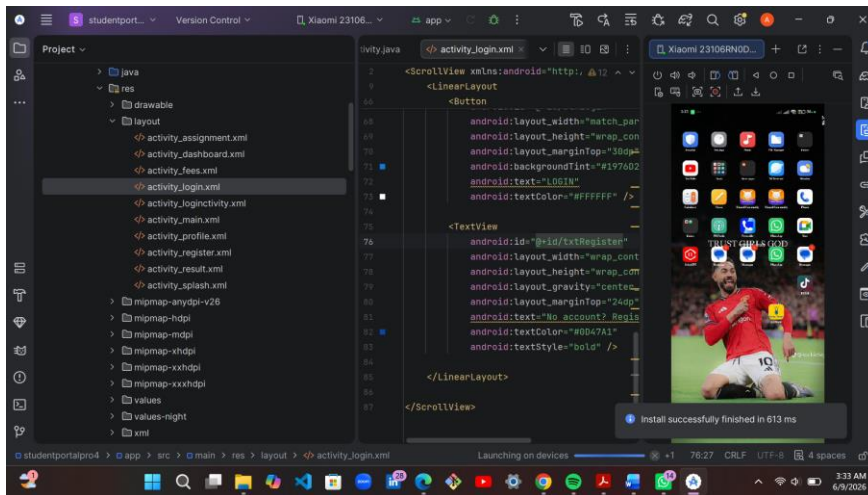


Figure 6.APP LAUNCHING

ON DEVICE

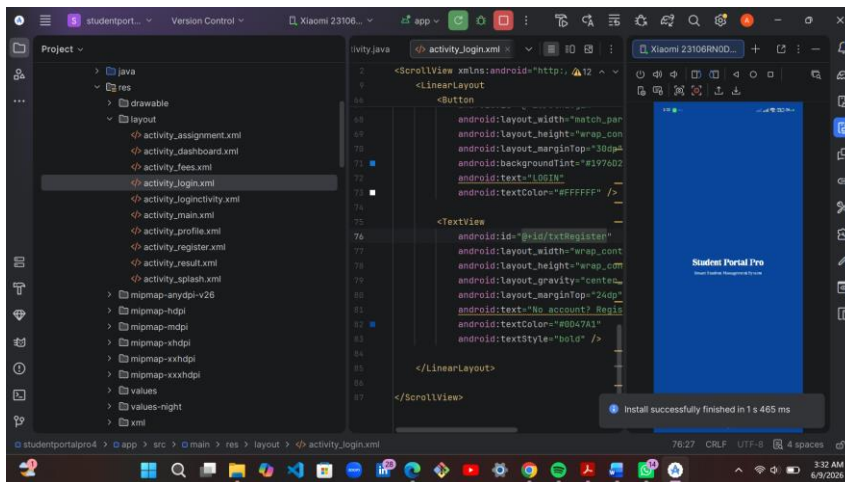


Figure 7.SUCCESSFULL

INSTALLATION

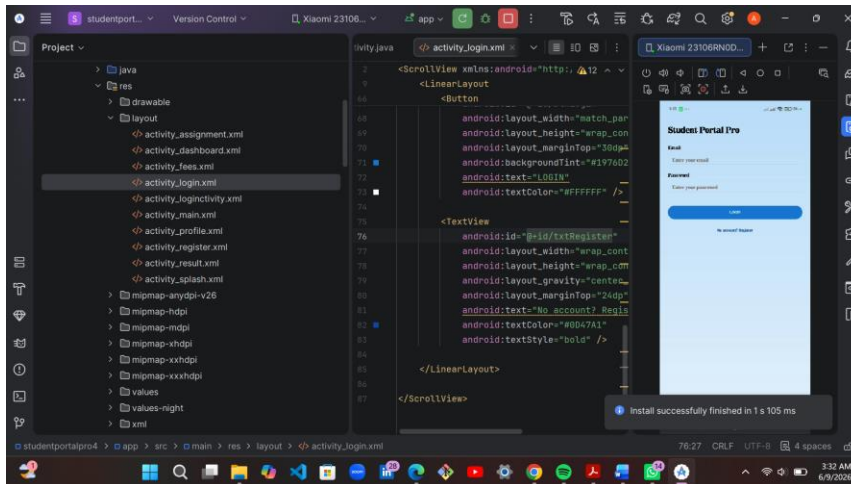


Figure 8.LOGIN SCREEN

ON EMULATOR VIEW

Tools/Software Used

- Android Studio
- XML
- Constraint Layout

Personal Reflection

This week I learned how to design user interfaces using XML in Android Studio. I also improved my understanding of layouts and positioning of UI components.

Week 3: User Interface Design

Practical Activities Undertaken

- Created dashboard screen for users
- Developed navigation between activities
- Added buttons for “Report Lost Item” and “Report Found Item”
- Improved UI design for better user experience

Dashboard screen

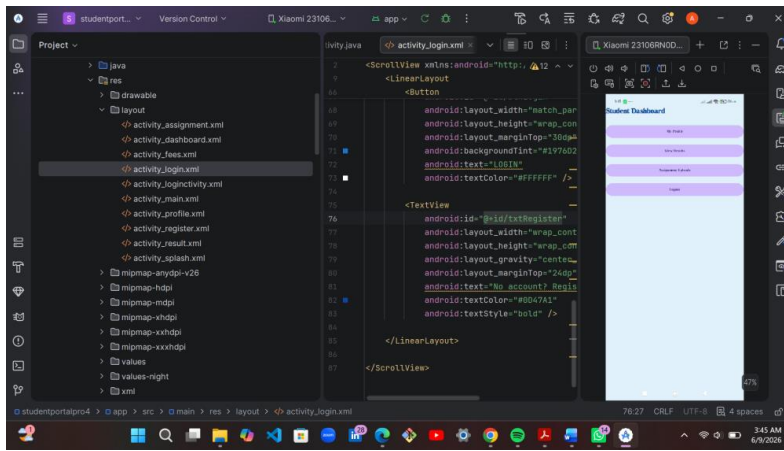


Figure 9. DASHBOARD SCREENS

Navigation between pages

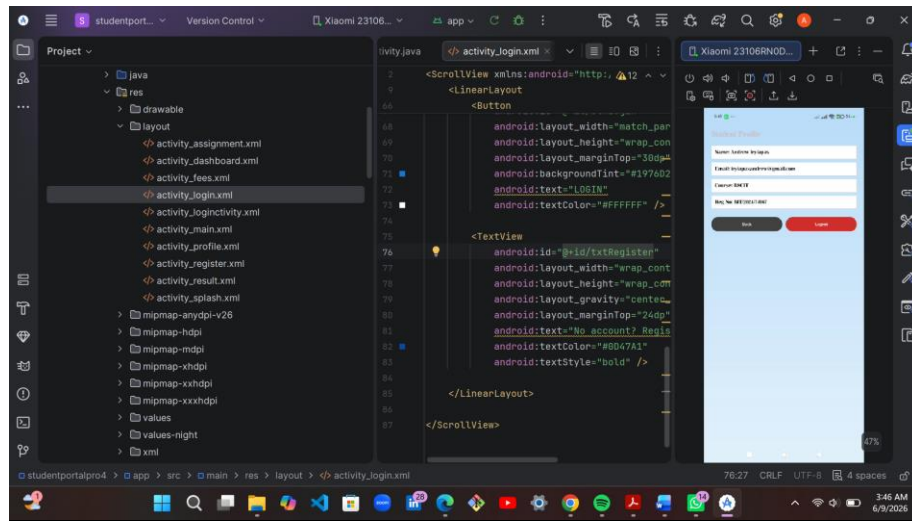


Figure 10.PROFILE BUTTON

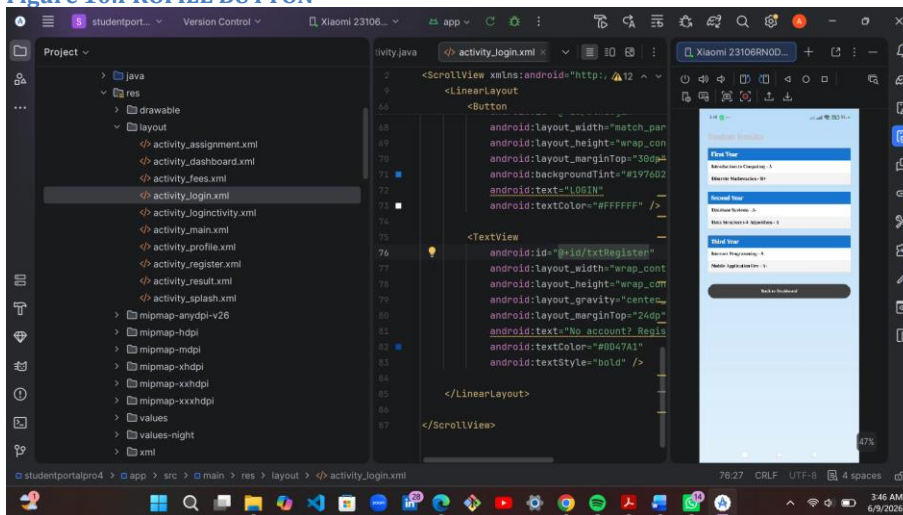


Figure 11.RESULTS BUTTON

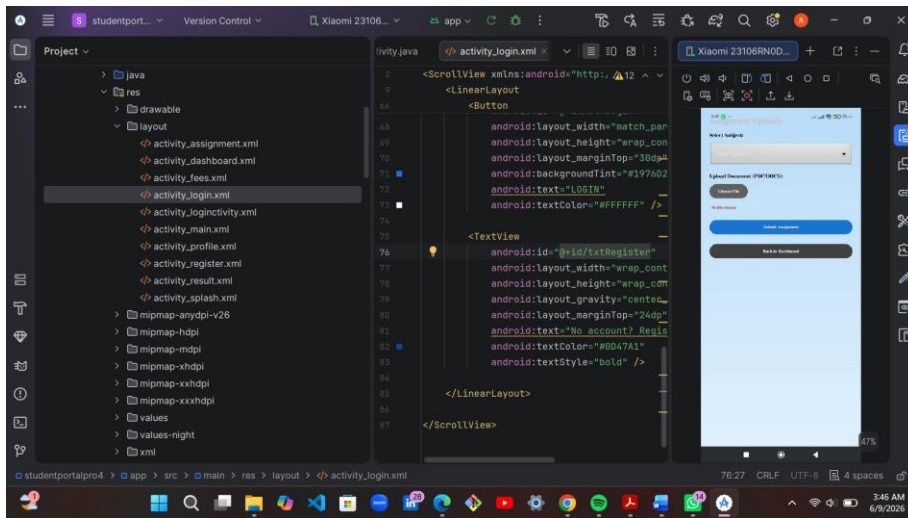
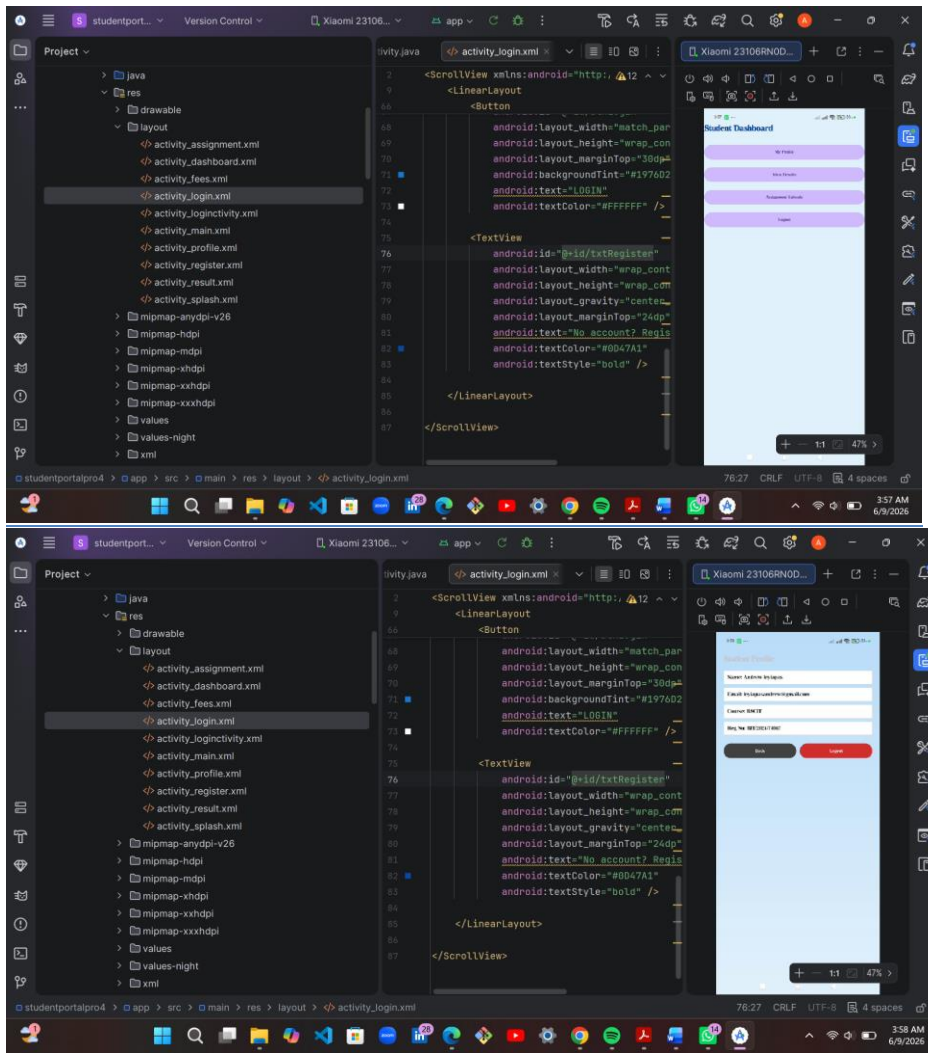


Figure 12.ASSIGNMENT BUTTON

Button interaction



Figure

13.DASHBOARD SCREENS &PROFILE BUTTONS&PROFILE SCREENS

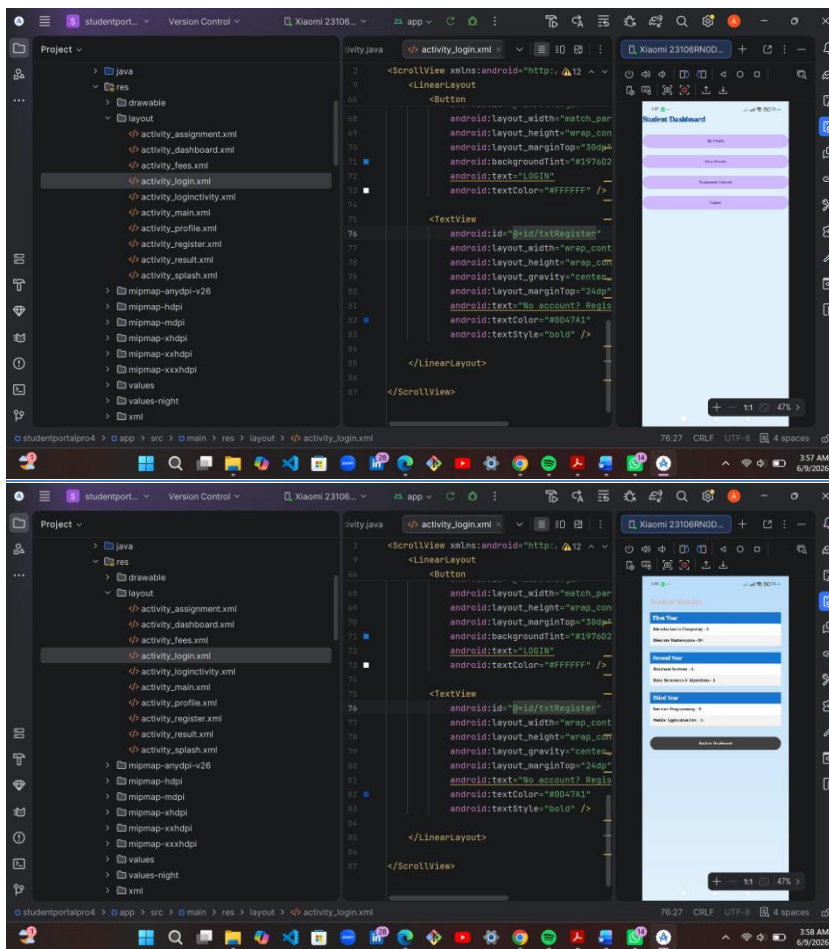
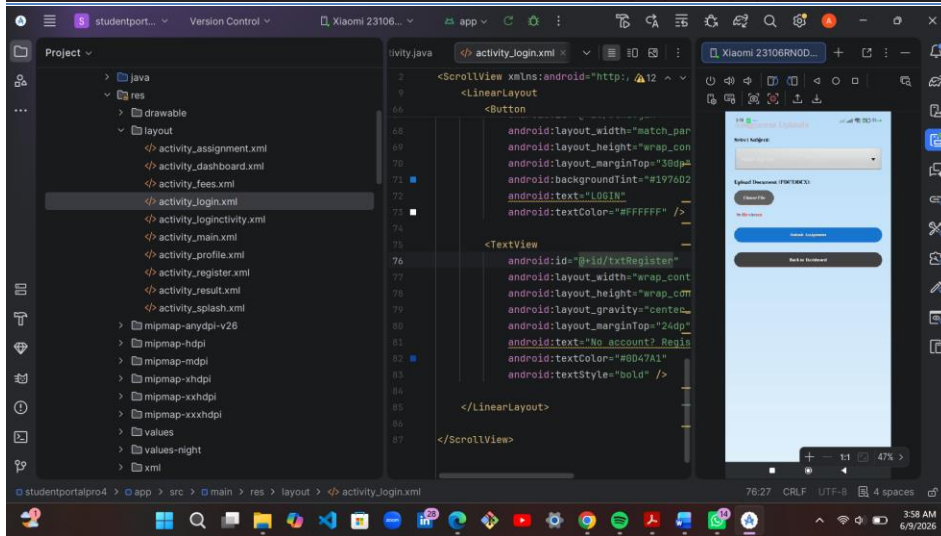
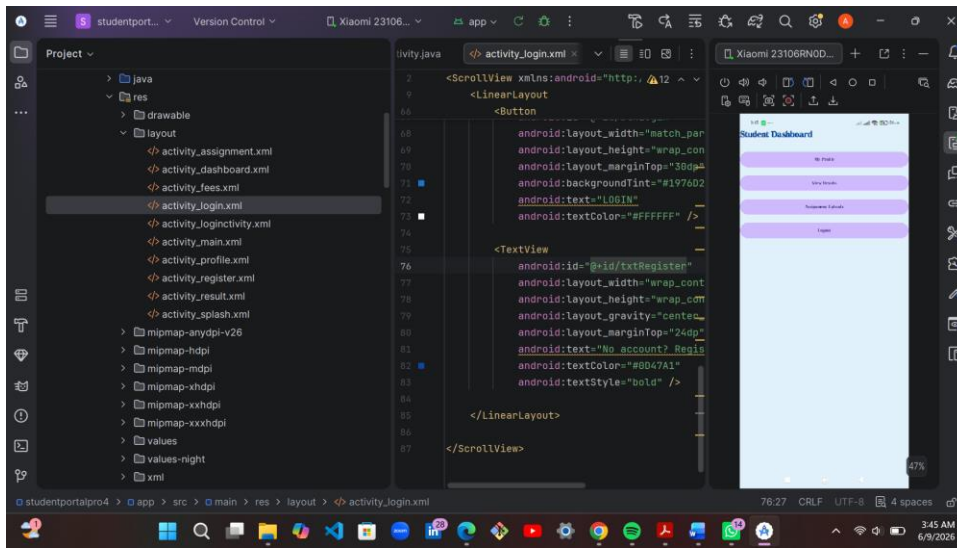


Figure 14.DASHBOARD

SCREENS&RESULTS BUTTON&RESULTS SCREENS



Figure

15.DASHBOARD SCREEN&ASSIGNMENT BUTTON&ASSIGNMENT SCREEN

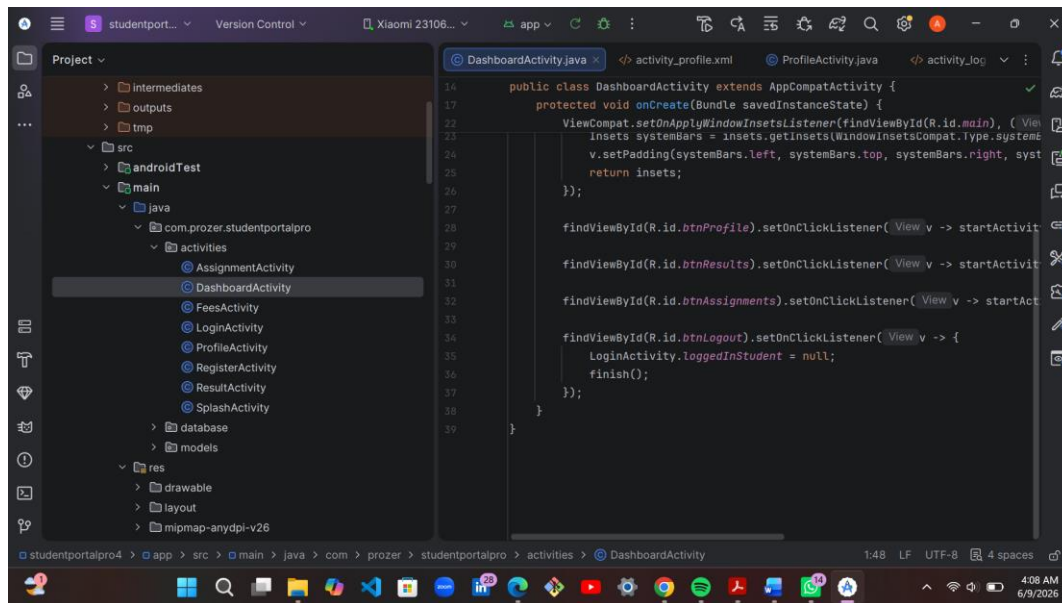
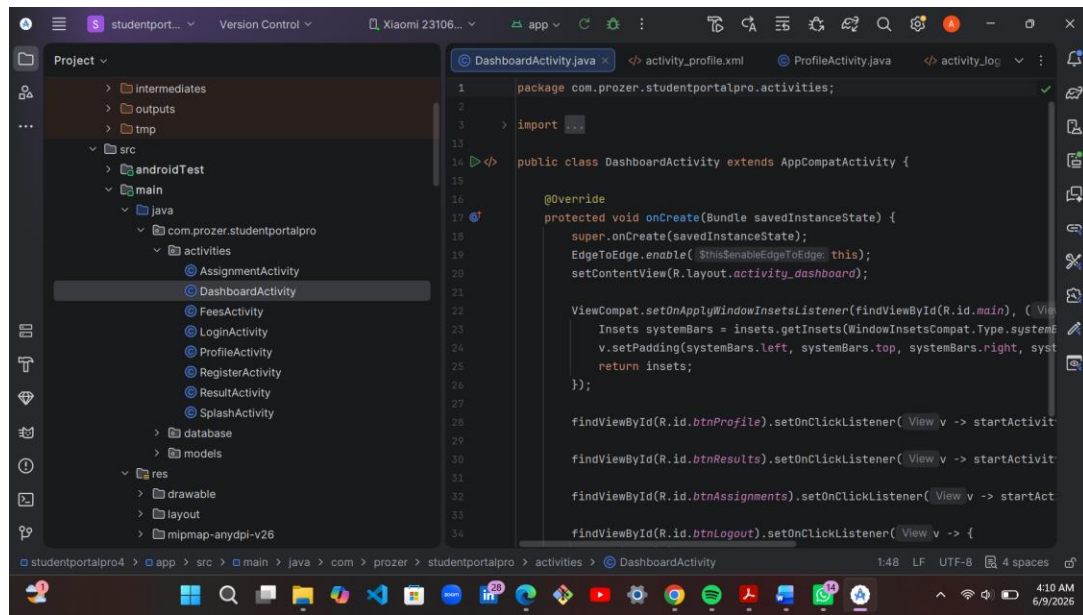


Figure 16. JAVA CODE IMPLEMENTING BUTTON CLICK

Personal Reflection

This week I learned how to connect multiple screens in an Android application. I also improved my understanding of navigation between activities and user interaction through buttons.

Week 4: Event Handling and User Interaction

Same as week 1

Week 5: Data Management

Same as week 1

Week 6: Networking and APIs

Same as week 1

Week 7: Application Architecture

Same as week 1

Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: _____

Student Signature: _____

Date: _____

Lecturer Name: _____

Lecturer Remarks: _____

Date: _____